

Insulation Materials and Where Heat Escapes

What each material is, where it belongs, and the one place heat leaves a well-insulated house anyway.

Loose fill	Blown-in material, best suited to insulating unconditioned attics, where it can simply be poured over an irregular floor.
Flexible insulation	Manufactured in both batts and blankets. Batts come cut to length, blankets in a roll.
Cellulose	Made of 80 to 85 percent recycled newspaper, with fire and pest retardants added.
Closed-cell foam	An oil-based, high density hard foam, 1.25 to 2 pounds per cubic foot. Sealed bubbles mean high R-value per inch and it acts as its own vapor barrier.
Open-cell foam	Light, soft and vapor-permeable. The density figure is what separates the two.
Urea-formaldehyde foam	Was taken off the market, but is no longer considered a health threat. Note the two halves of that answer — it left the market, and the concern has since been withdrawn.

YOU HAVE MISSED THIS 2 TIMES

Urea-formaldehyde foam insulation:

Was taken off the market but is no longer considered a health threat.

Rigid insulation and fire

YOU HAVE MISSED THIS 2 TIMES

To ensure that rigid insulation is fire-resistant:

It should be covered with at least 1/2 inch gypsum wallboard.

Foam plastic burns and gives off smoke, so it may not be left exposed inside a building. A half inch of gypsum is the thermal barrier that buys escape time.

Measuring what is already there

YOU HAVE MISSED THIS

The best way to positively determine how much insulation is in a finished wall is to:

Make a small hole in the wall and measure it.

Every other method is inference. The word doing the work in that question is "positively" — it is asking how to know for certain, not how to estimate.

Where the heat actually goes

YOU HAVE MISSED THIS 2 TIMES

In a well-insulated house that has storm windows, the greatest source of heat loss is through:

Air leakage at moveable joints.

Once the insulation and the glazing are dealt with, what is left is air simply moving through gaps — around opening windows, doors and hatches. Insulation slows conduction; only sealing stops leakage, and leakage is what remains.

YOU HAVE MISSED THIS

Heat flowing through exterior wall studs is called:

Thermal bridging.

A stud conducts far better than the insulation on either side of it, so heat takes the shortcut through the wood. Continuous exterior insulation is what breaks the bridge.

Say it back

Cover the page. Answer each one out loud.

1. Where does loose fill belong, and what forms does flexible insulation come in?
2. What is cellulose made of?
3. Open-cell against closed-cell foam — which is which, and what separates them?
4. State both halves of the urea-formaldehyde answer.
5. How is rigid insulation made fire-resistant?
6. How do you positively determine insulation in a finished wall?
7. Greatest source of heat loss in a well-insulated house with storm windows?
8. What is thermal bridging, and what stops it?